

$$S(T_n) = \frac{DF(T_0) - DF(T_n)}{\sum_1^n \delta_i DF(T_i)}$$

$$x^+ = \max(x, 0)$$

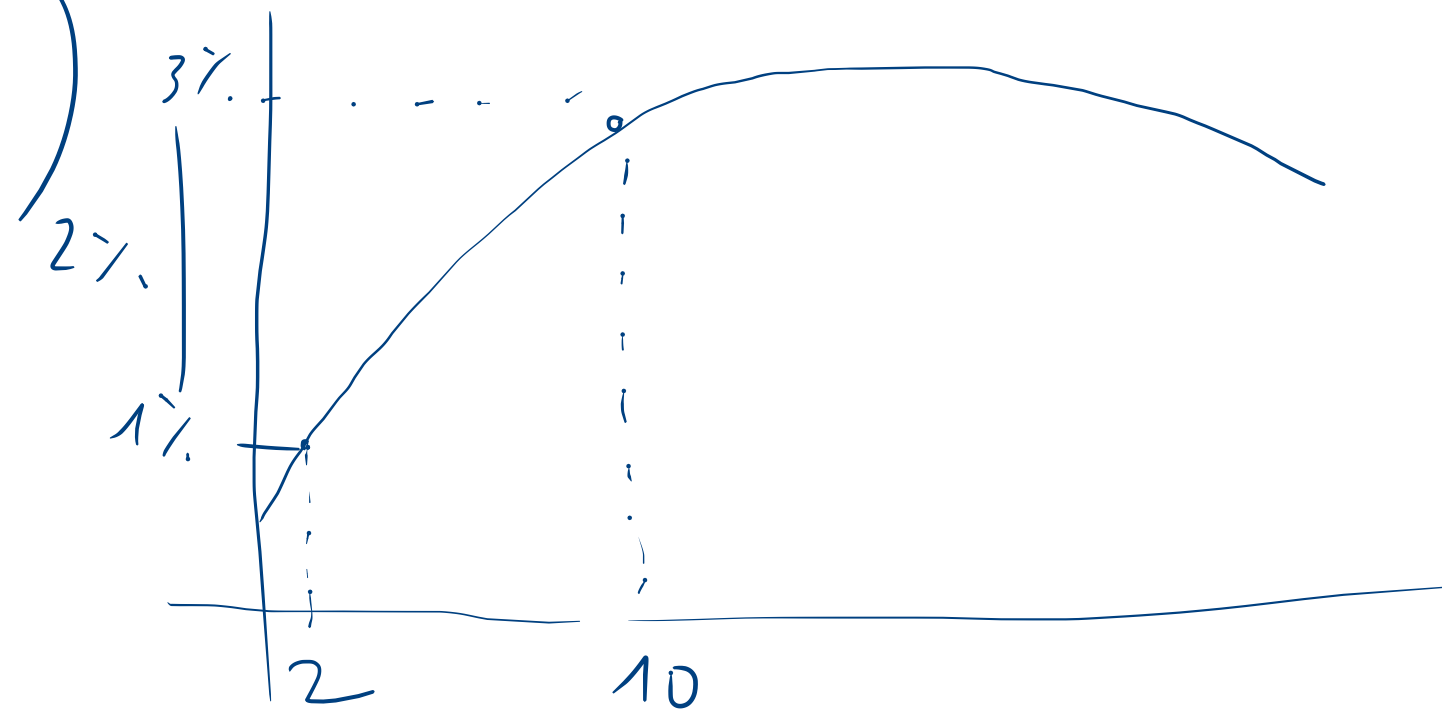
$$DF(T_n) + S(T_n) \sum_1^n \delta_i DF(T_i) = DF(T_0)$$

$$DF(T_n) (1 + \delta_n S(T_n)) = DF(T_0) - S(T_n) \sum_{i=1}^{n-1} \delta_i DF(T_i)$$

$$DF(T_n) = \frac{DF(T_0) - S(T_n) \sum_{i=1}^{n-1} \delta_i DF(T_i)}{1 + \delta_n S(T_n)}$$

$$C_0 = \mathbb{E}^Q \left(e^{-\int_0^T r_s ds} (S_T - K)^+ \right)$$

$$e^{-rt} \mathbb{E}^Q ((S_T - K)^+)$$



Reverse Floater

$$C_{pn} = (7\% - \text{EURIBOR})^+$$

Corridor

$$C_{pn} = 9\% \times \frac{n}{N}$$

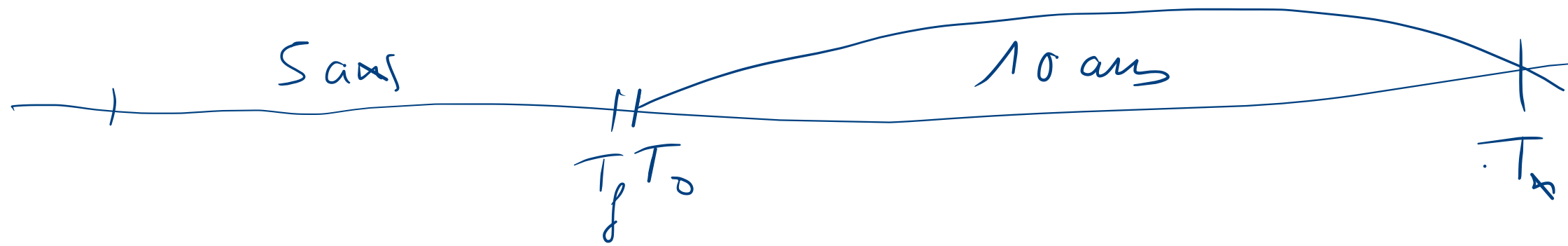
$\leftarrow$ nb jours où $EUR < 3,5\%$
 $\leftarrow$ nb total de BD

CMS spread

$$C_{pn} = \frac{5}{6} \times (\text{Taux Swap } 10y - \text{Taux Swap } 2y)^+$$

Parité CAP/Floor

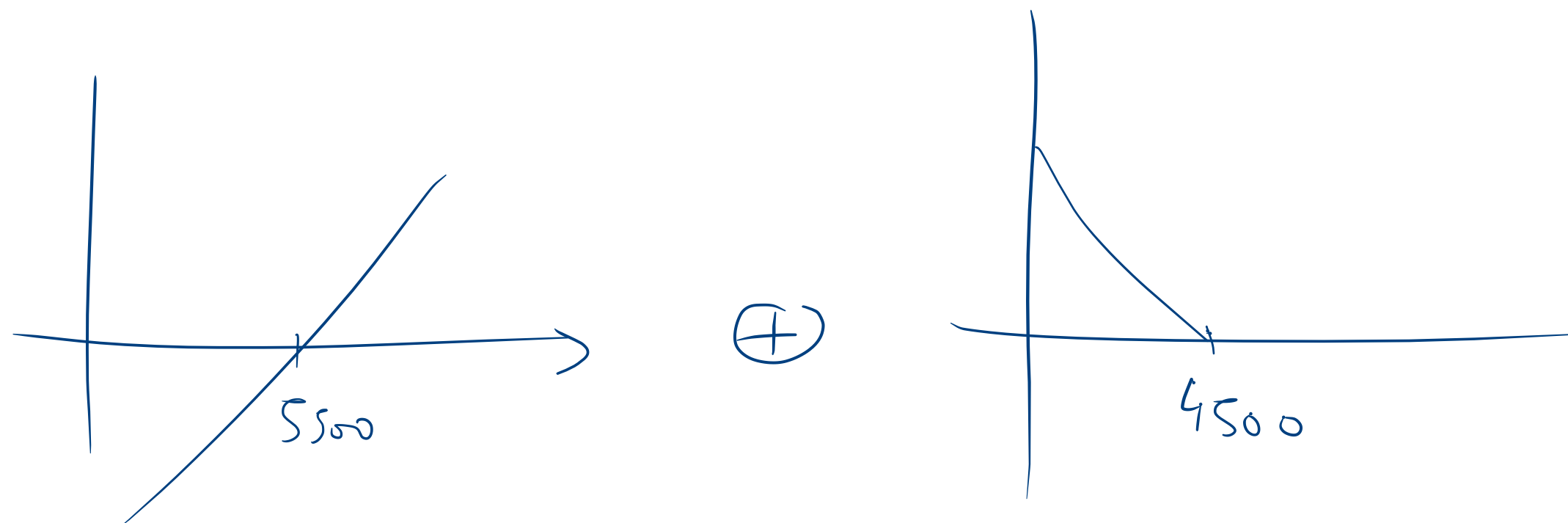
$$\underbrace{(EURIBOR - K)^+}_{CAP} - \underbrace{(K - EURIBOR)^+}_{FLOOR} = \underbrace{EURIBOR - K}_{SWAP}$$



$$PV_{\text{Swap}}(T_f) = \underbrace{LVL(T_f)}_{\sum \delta_i B(T_f, T_i)} (S(T_f) - K)$$

$$PV_{\text{Swaption}}(T_f) = LVL(T_f) (S(T_f) - K)^+$$

$$\max(S_t - K, C_t^+)$$



$\oplus$

$=$

